

Laboratory sessions

Laboratory assignment 7

High-level design of a signal processing system

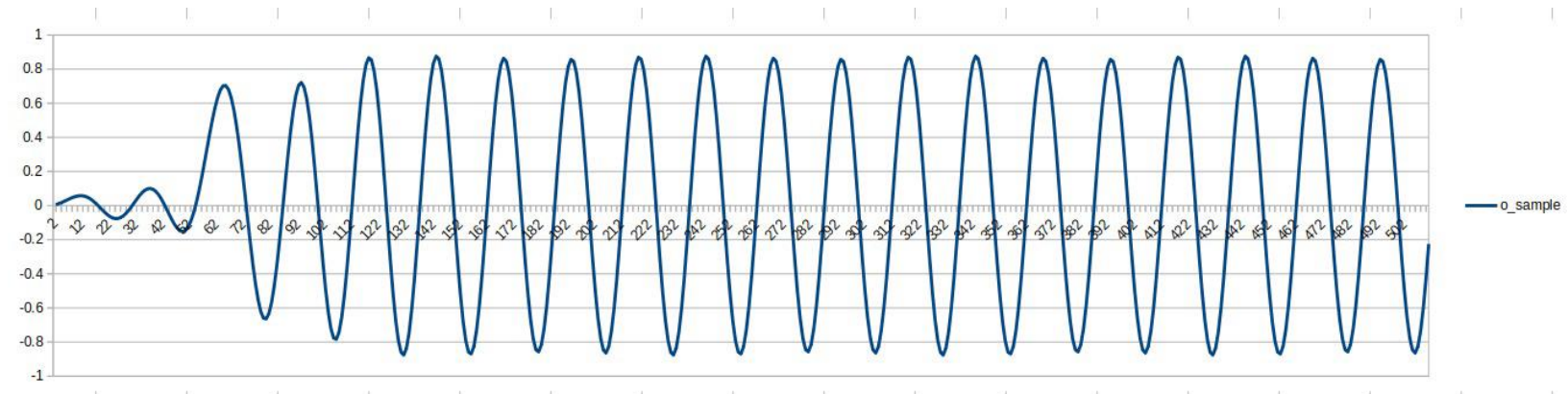


Goals

- ❑ Review an existing low-pass filter design in C++
- ❑ Adapt the low-pass filter design using algorithmic C bit-accurate data types
- ❑ Design and optimize a band-pass filter regulating the output amplitude with a Variable Gain Amplifier (VGA)
- ❑ Optional: Study the trade-off between resources and throughput with the Catapult synthesis tool

Part I – C++ low pass filter simulation

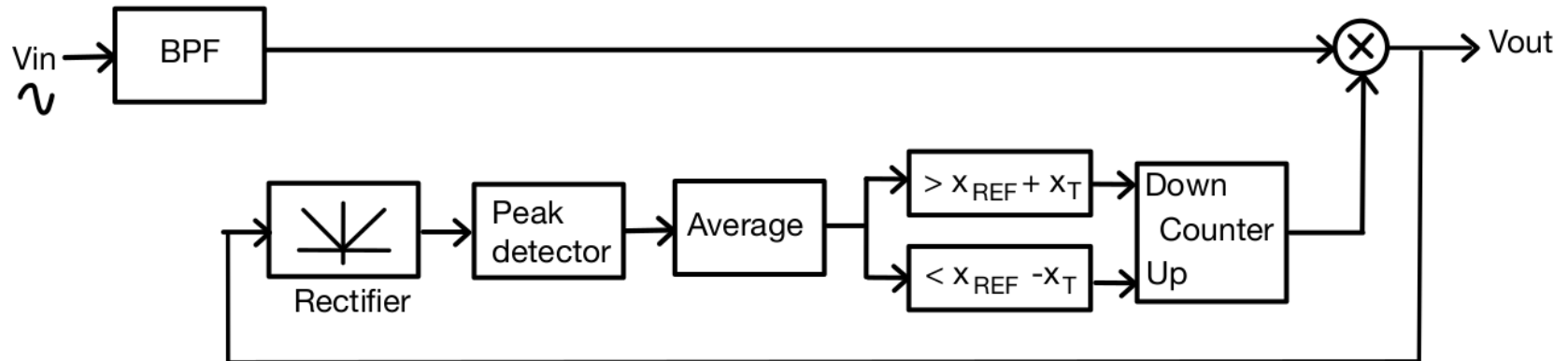
- ❑ 10 kHz sampling rate
- ❑ Band-pass at 400 Hz and stop-band (-40 dB) at 500 Hz
- ❑ Signal combinations to check:
 - 200 + 300 Hz, 300 + 400 Hz, 400 + 500 Hz, 500 + 600 Hz
- ❑ E.g. Response for the 400 + 500 Hz combination:



Part II – Bit accurate data types

- ❑ Change double types by:
 - `typedef ac_fixed<10,2,true> X_TYPE;`
 - `typedef ac_fixed<12,2,true> COEFF_TYPE;`
 - `typedef ac_fixed<12,4,true,AC_RND,AC_SAT_SYM> Y_TYPE;`
- ❑ Check the Low-pass filter behavior with new types
- ❑ Check the functionality of a band-pass filter behavior with new types
- ❑ Same combination of signals as Part I

Part III – Band-pass filter with Automatic Gain control



Lab report

❑ Should include

- Plots of all waveforms of Part I and Part II with comments.
- Part III: C++ source code with comments.
- Include design decisions, in particular the chosen parameters.
- Plots demonstrating the good operation of the Part III system.
- Conclusion with observations.